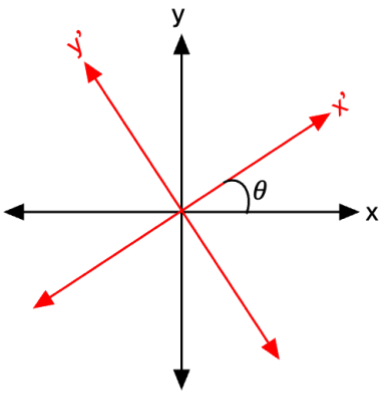
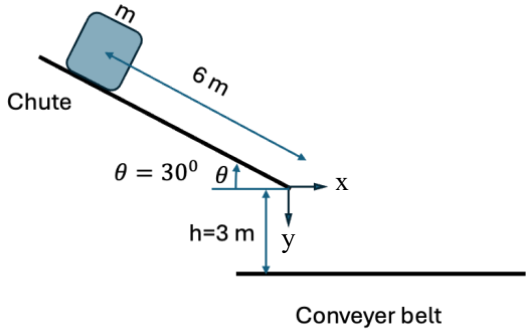
JK Lakshmipat University, Jaipur
Institute of Engineering and Technology
AS1108: Applied Physics Fall-2026
Assignment-1

1. Show that $A_x \cdot B_y - A_y \cdot B_x$ remains unchanged under the rotation of axis. Hint: See the figure below and use the information provided to resolve the vectors A and B in the x′-y′ frame of reference.



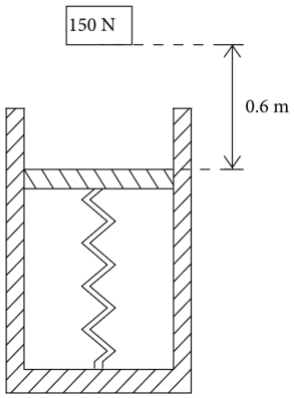
$$x' = x\cos\theta + y\sin\theta$$
$$y' = -x\sin\theta + y\cos\theta$$

2. A small box starts from rest and slides 6 m down a frictionless chute angled at 30° to the horizontal. At the end of the chute it drops onto a conveyer belt passing 3 m below. Where does it land on the conveyer belt and what is the speed with which it lands?

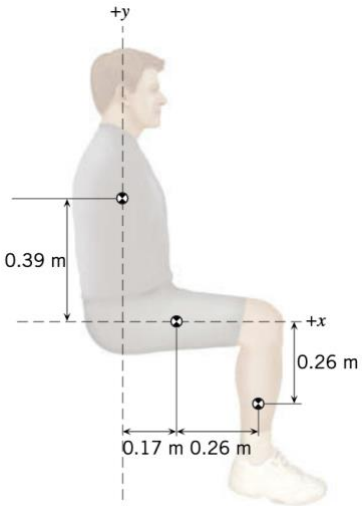


3. A circular turntable rotates counter-clockwise at a constant angular speed of 4 rad/s. An ant starts walking from the center of the turntable radially outwards with a constant radial speed of 0.05 m/s relative to the turntable. Determine the following when the ant is 0.2 m from the center:
   a. The radial and the transverse (in polar coordinates it is the "Ø" component) components of the velocity of the ant.
   b. The radial and the transverse components of the ants acceleration.

4. Consider a cylinder-piston arrangement where a spring is placed as shown in the figure below. A block of weight 150 N is allowed to fall on piston from height 0.6 m If the stiffness of spring is 1.20 N/mm, determine:
   a. the compression of spring by using the principle of conservation of energy.
   b. the maximum force exerted by the spring on the piston.
   c. the acceleration of the block just before it comes momentarily to rest.

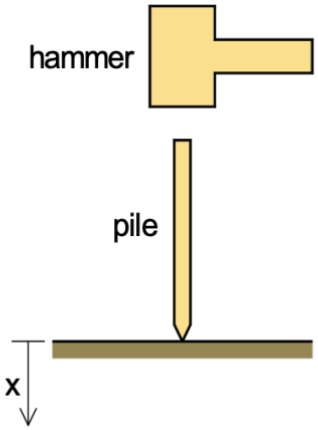


5. The drawing shows a human figure in a sitting position. For purposes of this problem, there are three parts to the figure, and the center of mass of each one is shown in the drawing. These parts are: (1) the torso, neck, and head (total mass = 41 kg) with a center of mass located on the y axis at a point 0.39 m above the origin, (2) the upper legs (mass = 17 kg) with a center of mass located on the x axis at a point 0.17 m to the right of the origin, and (3) the lower legs and feet (total mass = 9.9 kg) with a center of mass located 0.43 m to the right of and 0.26 m below the origin. Find the x and y coordinates of the center of mass of the human figure. Note that the mass of the arms and hands (approximately 12% of the whole-body mass) has been ignored to simplify the drawing.

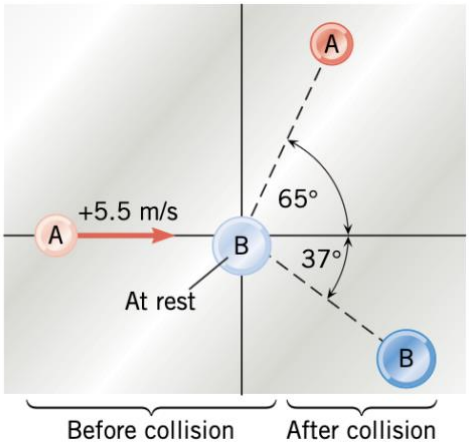


Now, if a cat of mass 10 kg jumps on to the man's lap, whose centre of mass is (0.50, 0.10), determine the new centre of mass of the man and the cat.

6. A pile of mass m=1500 kg is initially at rest. A hammer delivers an impulse of 4500 N.s to the pile, causing it to move downwards into the ground. As the pile penetrates the ground, it is subjected to its weight mg acting downwards and a resistive force acting upwards given by the expression $F_R = mg + 30000\,x\,v$, where x is the depth of penetration into the ground (m) as shown in the figure below and v is the downward velocity of the pile in m/s. Find the velocity of the pile immediately after the impact. Determine the maximum depth x to which the pile is driven by this single hammer strike.



7. During a game of carrom, a moving coin strikes another stationary coin near the center of the carrom board. The moving coin (Coin A) has a mass of 0.025 kg and initial speed of 5.5 m/s along the x-axis. The stationary coin (Coin B) has a mass of 0.050 kg. Because, the collision is not head-on, the two coins move away in the direction shown in the figure after the collision. Find the final speeds of coin A and coin B.
   [NOTE: Before substituting any numerical values, obtain the symbolic expression for the final speeds of the coins described in this collision problem. Let the initial velocity of coin A be u and final velocities be v_A and v_B. Let the coins move away at angles α and β after the collision. Use conservation of momentum in x and y directions.]



8. The motor in an electric saw brings the circular blade from rest up to the rated angular velocity of 80.0 rev/s in 240.0 rev. If the moment of inertia of a blade is 1.41×10^{-3} kg.m^2. What net torque (assumed constant) must the motor apply to the blade?